

Hardware-In-the-Loop Simulation of a High Frequency Interleaved Converter based on a Low-Cost FPGA Platform

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Abstract—Hardware-In-the-Loop (HIL) simulation is commonly used in multiple applications to validate control systems. Nevertheless, its application to high frequency power converters is particularly challenging due to the low time step required. This paper reports the setup of an HIL real time simulation bench, used to test the closed-loop control of a high frequency converter. A single phase interleaved bridgeless totem pole PFC running at a switching frequency of 100 kHz is simulated and driven by a C2000 microcontroller unit. The solver is based on a state-space representation and precomputed matrices, and implemented on a low-cost Artix-7 FPGA. Running with a time step as low as 50 ns, the simulation shows a good fidelity both in open-loop and closed-loop control. A detailed breakdown of error contributions is provided and analyzed in the frequency domain. Dynamic response analysis in closed-loop operation is also carried out comparing HIL results with an offline simulation and a small-signal model. The proposed bench shows the precomputed matrices method is reliable and accurate, with an excellent performance to platform cost ratio.

Index Terms—HIL, real-time simulation, time step, FPGA, high frequency, power converters

I. INTRODUCTION

In many engineering fields, HIL simulation is used to test control systems to accelerate development cycle, increasing flexibility of tested scenarios, along with safety, and no risk of material damages on the controlled system, as it is simulated. In power electronics, HIL simulation is both useful and challenging to develop, particularly with wide band-gap transistors that allow higher operation frequencies. Indeed, a simulation time step significantly lower than the switching period (at least 20 times lower and then going under the μ s) is required to ensure simulation fidelity. Moreover, a low latency is needed to accurately simulate the system dynamics, which is particularly fast in power converters.

In this context, a number of solutions have been proposed to meet this need. Commercial Real-Time Simulators (RTS) are proposed by several companies and are based on either FPGA or CPU platforms. The achieved time step is typically limited to around 200 ns on the highest grade platforms [1]. The solver is often based on associated discrete circuit modelling initially proposed in [2], which has the drawbacks of introducing unre-

alistic oscillations and having limited accuracy. Compensation methods to alleviate this issue have however been proposed in [3] [4]. In the literature, several methods have been proposed, like detailed representation of switch transients with increased complexity [5] [6] [7], and ideal or resistive switches models that model the converter with multiple matrices depending on switches configuration. This last method is similar to offline simulation software such as *PLECS*, and requires online matrix computation as proposed in [8], or offline precomputation and storage of all the matrices [9] [10] [11]. Tearing techniques have also been proposed, breaking the discretized numerical model into parts to decrease the computational cost [12].

The approach proposed in this paper is to implement a high frequency Power Factor Correction (PFC) converter simulation using precomputed matrices, as it can achieve low time steps with a high accuracy, despite needing more logic resources. A simple converter configurations analysis is conducted to reduce the number of matrices to store. Matrices are also simplified with methods similar to those used with sparse matrices. The solver is implemented on a Xilinx Artix-7 FPGA to demonstrate the feasibility of high frequency converter real-time simulation using inexpensive hardware. The simulator performance is assessed with average error measurement and frequency-domain analysis. The dynamic response fidelity is also analysed and compared to a small-signal model, and closed-loop testing is conducted with the converter driven by a C2000 microcontroller unit [13].

II. TOTEM-POLE PFC CONVERTER MODELLING

The converter under study is a GaN-based bridgeless totem-pole PFC converter, whose electrical diagram is given on Fig. 1. It is designed to operate at a 100 kHz switching frequency on a 230 V / 50 Hz grid, and can deliver up to 3.3 kW. This topology is generally used in grid interconnection of energy storage systems and electric vehicles chargers. The AC input voltage is rectified by the low frequency leg (leg 0) switched at grid frequency, while the high frequency legs (legs 1, 2, 3) are interleaved to reduce input current ripple and Total Harmonic Distortion (THD).

The control scheme is based on nested current and voltage loops, regulating the output voltage at a nominal 380 V. The fidelity analysis of the real-time model will focus on the current loop, as it is way faster than the voltage loop. The state-space model of the converter is established, taking into account the multiple configurations of the converter depending on the switches states. To compare the frequency response of the HIL simulator with the real converter, the small-signal model will be derived further in Section VI-C.

The real-time model of the converter is based on a large-signal state-space representation of the converter. This model is closely related to the states of the switches, whose determination needs to be done by the solver, as explained in Section III-B. Applying Kirchhoff laws on the electrical circuit in the different configurations, the state-space representation is established as given in (1), G_x being the control signal of the switch x . The state vector X contains the inductor currents and the capacitor voltage, while the input vector U contains the input grid voltage and the output load current. In the output vector Y , we can recover the total input current by summing the inductor currents.

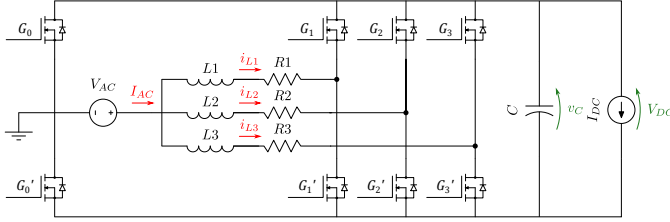


Fig. 1. Circuit diagram of the interleaved totem-pole PFC converter

$$\begin{cases} \dot{X} = M_r A_r X + M_r B_r U \\ Y = C X \end{cases} \quad (1)$$

$$X = \begin{bmatrix} i_{L1} \\ i_{L2} \\ i_{L3} \\ v_C \end{bmatrix} \quad U = \begin{bmatrix} v_{AC} \\ i_{DC} \end{bmatrix} \quad Y = \begin{bmatrix} i_{L1} \\ i_{L2} \\ i_{L3} \\ v_C \\ i_{AC} \end{bmatrix} \quad (2)$$

$$A_r = \begin{bmatrix} -r_1 & 0 & 0 & G_0 - G_1 \\ 0 & -r_2 & 0 & G_0 - G_2 \\ 0 & 0 & -r_3 & G_0 - G_3 \\ G_1 - G_0 & G_2 - G_0 & G_3 - G_0 & 0 \end{bmatrix} \quad (3)$$

$$M_r = \begin{bmatrix} L_1^{-1} & 0 & 0 & 0 \\ 0 & L_2^{-1} & 0 & 0 \\ 0 & 0 & L_3^{-1} & 0 \\ 0 & 0 & 0 & C^{-1} \end{bmatrix} \quad (4)$$

$$B_r = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (5)$$

III. REAL-TIME SOLVER

A. Model discretization

The state space model of the converter is discretized using a first order backward Euler scheme (BE). This method is more stable (A-stable) and accurate than the forward Euler scheme, while being less computationally demanding than higher order methods [14]. It is also particularly suitable to deal with switching discontinuities, unlike trapezoidal scheme which introduces unrealistic oscillations around them (despite being generally more accurate) [15]. Applying the BE scheme to the state-space model, the discretized system equations are obtained in (6).

$$\begin{cases} v = \mathcal{M}^{-1} b \\ y = C v \end{cases} \quad (6)$$

$$\begin{aligned} v &= X_{k+1} && \text{state vector} \\ y &= Y_{k+1} && \text{output vector} \\ b &= X_k + h M_r B_r U_{k+1} && \text{source vector} \\ \mathcal{M} &= I - h M_r A_r && \text{discrete state matrix} \end{aligned}$$

The source vector b contains both the history terms (state vector at the previous time step) and the input terms (from the input vector at the current time step). The M matrix called the discrete state matrix depends on the converter configuration (related to the switches states). Therefore, it is inverted and stored in all its configurations to be used by the solver.

B. Switches states determination and matrix selection

The solver is implemented on the FPGA platform as described in Fig. 2. The time step value is 50 ns. It accounts for 200 steps per switching period at 100 kHz, which is above the standards (20 to 100 steps per period).

To select the appropriate matrix, the solver needs to determine switches states K_X , based on control signals G_X and the state vector at the previous time step. A major concern in real-time simulation of power electronics converters is diode state calculation, due to unpredictability of the switching instant. Several methods have been proposed to determine diode state. For example, a method using current mapping with diode states has been proposed in [14]. As diodes conduct mainly during dead times in the present application, a state determination based on the current polarity is sufficient.

To reduce the number of matrices to store, some configurations are excluded or redirected to equivalent states, ending up with 15 matrices. The states where the cells are short-circuited are excluded because they are forbidden states by driver design, while the states where both switches of a cell are open are redirected to the conduction of one of the body diodes, depending on their state. Moreover, ones and zeros are identified and simplified to reduce DSP consumption on the FPGA.

Once the solver has determined the states of the switches K_X , the appropriate matrix is retrieved from a look-up table. Simultaneously, the solver updates the b vector using inputs and history terms. The solver finally computes the new state and output vector using matrix-vector products.

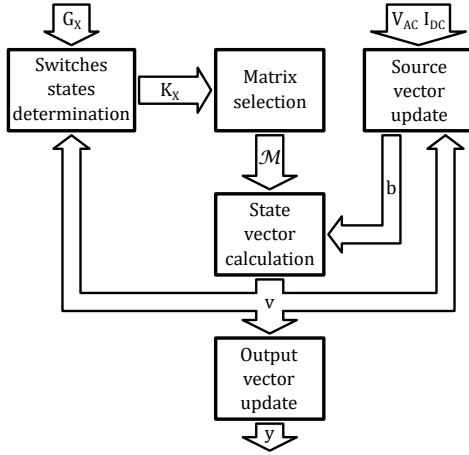


Fig. 2. Proposed solver structure

IV. IMPLEMENTATION

Since real-time simulation time step on CPU is generally limited above the μs , thereby limiting switching frequency to about 10 kHz, FPGA platforms are best suited to achieve the 50 ns required in our application. The solver is implemented on a low-cost Xilinx Artix-7 XC7A100T, whose logic specifications are given in Tab. I.

A. Number representation

The solver uses a 32-bit fixed-point binary format, with 20 bits for the fractional part. While floating-point arithmetic allows a wide range of values and a higher precision, it requires a dedicated Floating-Point Unit (FPU) whose implementation on an FPGA is complex and computationally demanding. The fixed-point format is robust and thrifty, but special attention must be paid to scaling needed to increase precision while avoiding overflows, as data range is much tighter. The solver uses a binary scaling factor of $\frac{1}{2^{20}}$, which is less intuitive than decimal but particularly convenient for scaling and offsetting (required by the interfaces) that can be done by simple bit shifts and selections. Considering this formatting, solver data can be written as in (7). It ranges from -2048 to 2048, with an accuracy of $5 \cdot 10^{-7}$.

$$x = c_{31} 2^{11} + \dots + c_{21} 2^1 + c_{20} + c_{19} 2^{-1} + \dots + c_0 2^{-20} \quad (7)$$

B. Interfaces

The real-time simulation is interfaced with the controller through digital and analog signals, as shown in Fig. 3. Input voltage and output current sources are provided by an external function generator and sampled by Analog-to-Digital Converters (ADCs) on the FPGA. The input voltage is also provided to the MCU for current reference synchronization and voltage feedforward. Digital Pulse Width Modulation (PWM) inputs coming from the MCU are sampled by the FPGA at each time step. The simulator outputs (input current and output voltage) are converted by Digital-to-Analog Converters (DACs) to be read by the MCU in closed-loop operation. The inductor currents are also provided by the simulator. Scaling is applied

on all analog signals to use the full ranges of ADCs and DACs, increasing precision and signal-to-noise ratio. An offset is applied onto alternating signals in order to comply with the unipolar ADCs and DACs.

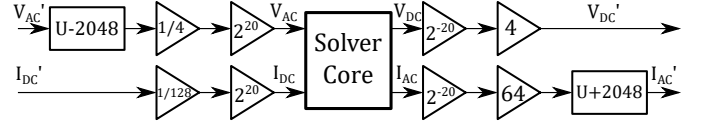


Fig. 3. Analog I/O formatting

C. FPGA

The simulator is described in VHDL and implemented in the FPGA using the *Vivado* software. The synthesis and implementation parameters are kept standards and no further optimization is conducted on the RTL design. The implementation achieved by *Vivado* is expected to meet timing with a worst negative slack of 5 ns. The resulting logic resources consumption is given in Tab. I. Though usually restrictive

TABLE I
FPGA RESOURCES

Resource	Utilization	Total	Percentage
LUT	1656	63400	2,61 %
FF	601	126800	0,47 %
DSP	48	240	20 %
IO	25	210	11,90 %
BUFG	3	32	9,38 %
MMCM	1	6	16,67 %

with precomputed matrices, the LUT consumption is kept very low thanks to matrices number reduction. The most restrictive resources are DSPs due to matrix-vector products in the solver and IOs that limits the use of parallel DACs.

V. SIMULATOR PERFORMANCE

A. Error estimation

In order to estimate the fidelity of the HIL simulator, the experimental results are compared with an offline simulation from *Matlab Simscape SPS*. This last is performed using a variable time step and a solver suitable for stiff differential equations (ode23tb), and can be considered as a good reference point as a first step to analyze the performance of the HIL simulator. The input current waveforms are given in Fig. 5. By averaging the current in both HIL and offline simulation on 1 ms, an error below 2% is measured, which is expected considering the solver and interface errors.

To complete this average estimation and to identify the different contributions to the error, we analyze the current waveforms both in HIL and offline simulation in the frequency domain using Fast Fourier Transform (FFT). The results are given in Fig. 4. Different groups of spikes are identified. At low frequency, the low amplitude harmonic content corresponds to the resonance frequency of the capacitor and inductors of the converter around 200 Hz. This is realistic and

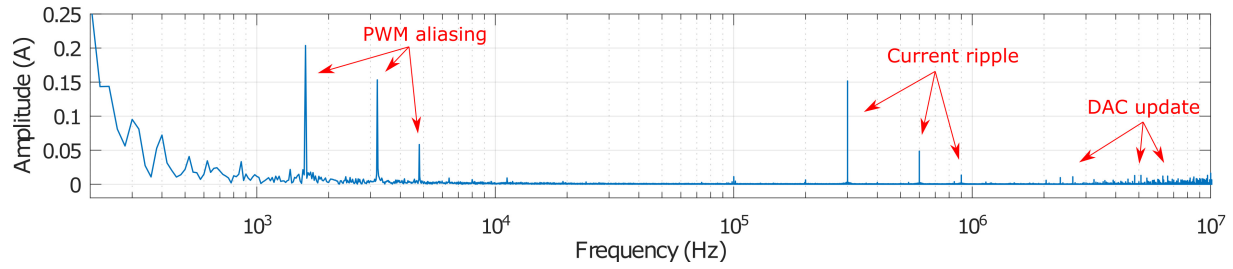


Fig. 4. FFT of inductor current

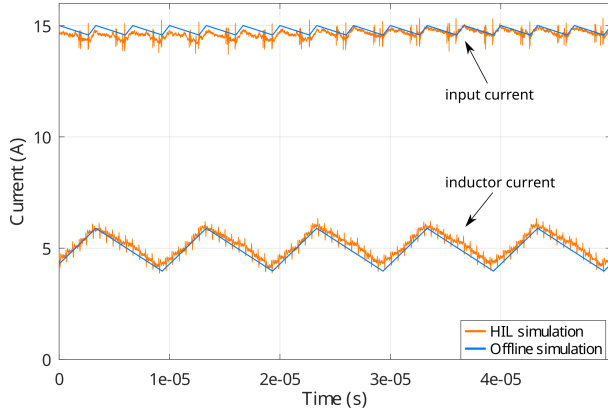


Fig. 5. Input current and output voltage in HIL and offline simulation

due to low frequency variations of the duty cycle. Frequency components at 1.6 kHz and its multiples are also identified. Those are unrealistic oscillations due to the aliasing effect on PWM signal sampling by the FPGA, which we discuss hereafter. Even though they are low amplitude regarding the DC value of 14 A, they are more significant than the realistic spikes at 300 kHz and its multiples which are the input current ripple components at 3 times the switching frequency (due to interleaving). Finally, spikes can be identified at high frequency above 3 MHz, and are due to DAC update events. Dynamic response fidelity of the current loop will be further investigated in Section VI.

B. Aliasing effect on PWM signal sampling

As shown in the frequency spectrum, significant harmonic content appears at a frequency of 1.6 kHz. This phenomenon is due to the aliasing effect on the sampled PWM inputs, which impacts the output current through the converter. Previous work has been done to extensively explain this phenomenon, particularly in [16]. As the FPGA and the MCU are not synchronized and the clock frequencies are not exact, the PWM signals are asynchronously sampled by the simulator at a finite frequency of 20 MHz, which introduces subharmonic oscillations, as shown in Fig. 6. The measured aliasing frequency of 1.6 kHz corresponds to a pattern repetition of 62 switching periods which is verified on the sampled PWM signals, thus confirming the origin of those harmonics. Further investigation on this phenomenon could be

considered to alleviate its impact on the output signals spectra.

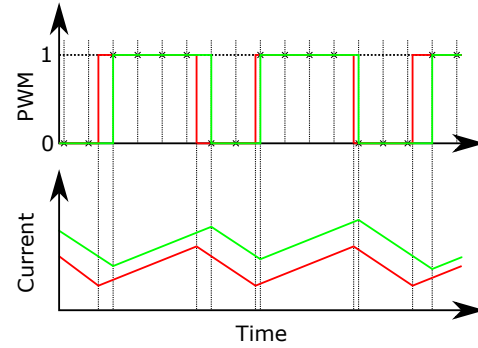


Fig. 6. Aliasing effect on PWM signal sampling

VI. LOOP PERFORMANCE

A. Physical prototype

The HIL simulation presented in this paper is meant to serve as a tool for validating the control software of a real, GaN-based totem-pole converter prototype shown in Fig. 7. In the scope of this work, this physical 3.3 kW converter has been used to provide an additional cross-comparison point, with results plotted and discussed in Section VI-D.

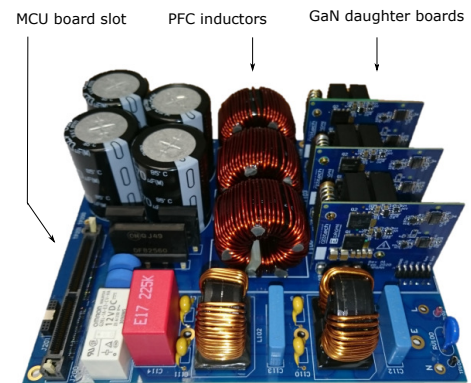


Fig. 7. Photograph of the physical totem-pole PFC prototype

B. Simulator latency characterization

While simulator latency does not have any impact in open-loop performance, it can be detrimental to dynamic response in

closed-loop operation. The multiple latency contributions are summarized in Fig. 8. The digital input synchronization and solver latency are both 50 ns, based on the simulation time step. The output interface latency is measured and results are depicted in Fig. 9. The latency is made up of :

- data serialization which represents 340 ns (17 cycles on a 50 MHz DAC) ;
- conversion time which is around 100 ns ;
- settling time which is 600 ns in the worst case (that is, a full range step).

The total latency of the simulator can therefore be estimated at about 1.14 μ s.

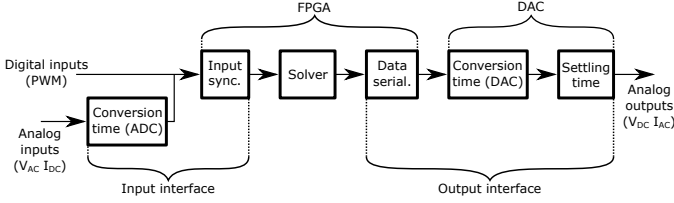


Fig. 8. Latency contributions

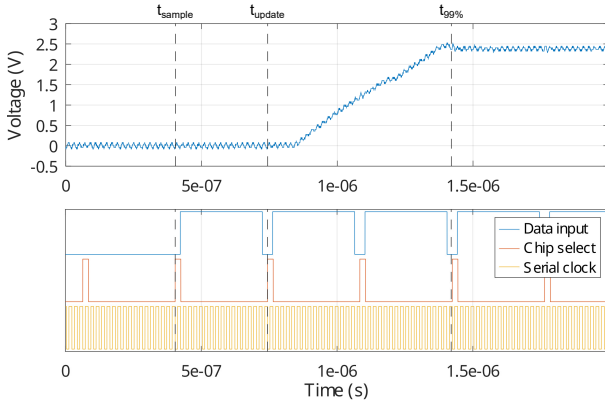


Fig. 9. DAC latency

C. Small-signal model

A small-signal model of the converter is established in order to serve as a reference for assessing the fidelity of the HIL simulator dynamic response. The transfer function of interest is from duty cycle to input current. Its expression is given in (8). The H block stands for the linearized small-signal model given in (9), while G_d models the PWM delay.

$$\frac{\tilde{i}_{AC}}{\tilde{d}}(s) = H(s)G_d(s) \quad (8)$$

$$\begin{cases} \tilde{\dot{X}} = MA\tilde{X} + MB\tilde{U} \\ \tilde{Y} = C\tilde{X} \end{cases} \quad (9)$$

$$\tilde{X} = \begin{bmatrix} \tilde{i}_L \\ \tilde{v}_C \end{bmatrix} \quad \tilde{U} = \tilde{d} \quad \tilde{Y} = \tilde{i}_{AC} \quad (10)$$

$$A = \begin{bmatrix} -r & -D \\ 3D & -\frac{I_{DC}}{V_C} \end{bmatrix} \quad (11)$$

$$M = \begin{bmatrix} L^{-1} & 0 \\ 0 & C^{-1} \end{bmatrix} \quad (12)$$

$$B = \begin{bmatrix} -V_C \\ 3I_L \end{bmatrix} \quad C = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \quad (13)$$

D. Dynamic response comparison

A small perturbation is added to the duty cycle input of the modulator, and the Bode diagram is drawn by sweeping the frequency and measuring the response on the current output of the simulator. The results are plotted alongside the small-signal model and the real converter on Fig. 10. The model including the simulator latency estimation of 1.14 μ s is also plotted. The results are consistent with both the model and the real converter. The HIL simulator latency impact is not as important as expected, because of the DAC settling time that was measured in the worst case. The impact of the simulator latency at the target loop bandwidth (which is around 7 kHz) is barely noticeable even against the real converter. One can reasonably expect a phase margin degradation less than 5°, which is fairly acceptable compared to the target phase margin of 45°. Also, as opposed to the HIL simulation measurements, and due to constraints related to the physical implementation, the real converter Bode plot has been assessed with poorer accuracy measurements, which results in some deviation with its small-signal model.

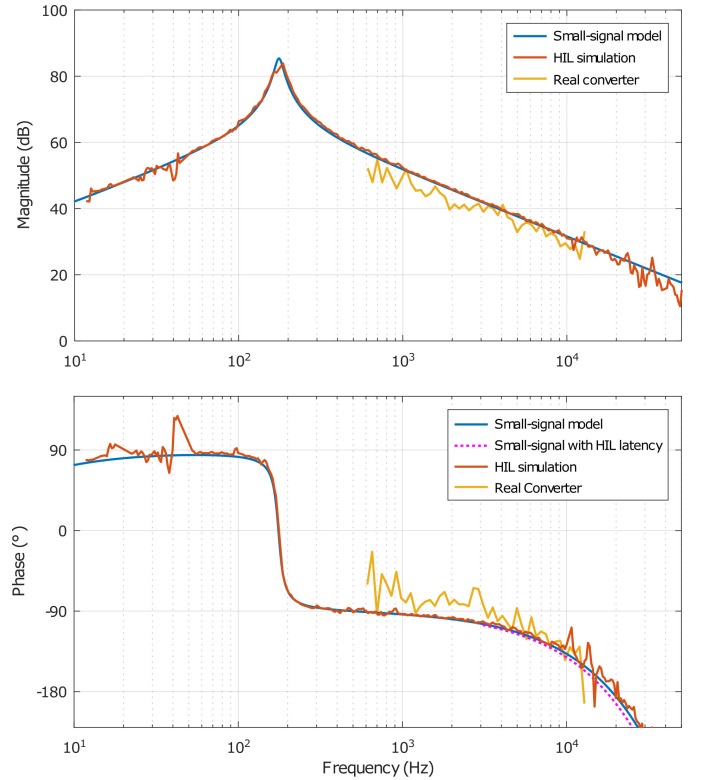


Fig. 10. Bode plot of the experimental results vs the small-signal model

The behavior of the simulator is verified in operation, as shown in the time-domain waveform capture in Fig. 11. As expected, the voltage output is regulated at 380 V DC, and

input current is around 14 A RMS with very low harmonic distortion. Regarding inductor currents, the imbalance due to phase interleaving is observed as in the real converter, meaning that the simulator can also be used to test current balancing control strategies.

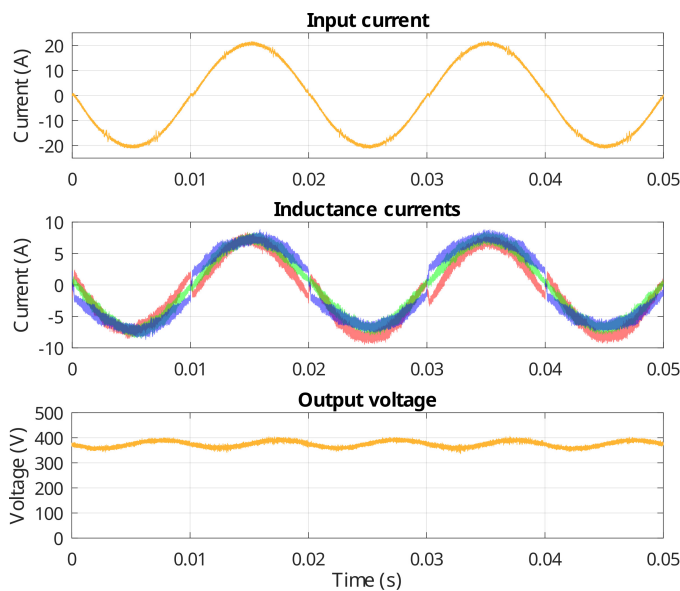


Fig. 11. HIL simulator waveforms in AC/DC operation with output voltage control

VII. CONCLUSION

A real-time simulator for a high frequency interleaved PFC converter is presented in this paper, allowing HIL testing of control strategies. The converter is modelled by a discretized state-space representation and implemented on FPGA with the precomputed matrices method. Measurements show a good average accuracy of 2% and a dynamic response very close to the theoretical model, thereby demonstrating the feasibility of HIL testing of high frequency converters controllers on low-cost FPGA platforms. Future work could contribute to further reduce the total latency of the simulator, as well as investigate methods to mitigate the aliasing effect on the sampling of PWM signals.

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